Xianfei Wang

xwanglk@connect.ust.hk | github.com/xianfei

Research interests: Human-AI interaction | Multimodal interactive systems | Spatial interaction and XR

EDUCATION

The Hong Kong University of Science and Technology *Hong Kong*
MPhil in Computer Science and Engineering *Sep 2025 - Jun 2027*

- Supervisor: Kani Chen | Expected graduation: June 2027

Beijing Information Science and Technology University *Beijing, China*
BEng in Software Engineering *Sep 2021 - Jun 2022*

- Completed final-year study and thesis at BISTU; undergraduate thesis supervisor: Wenfeng Song.

Beijing University of Posts and Telecommunications *Beijing, China*
Undergraduate Coursework in Software Engineering *Sep 2018 – Aug 2021*

- Joint Training Program with BISTU; completed undergraduate coursework at BUPT.

PUBLICATIONS

2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)
Poster paper
Wenfeng Song†, **Xianfei Wang**, Yang Gao, Aimin Hao, and Xia Hou. "Real-time Expressive Avatar Animation Generation based on Monocular Videos." 2022, pp. 429-434. DOI

2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)
Poster paper
Wenfeng Song†, Xinyu Zhang, Yang Gao, Yifan Luo, Haoxiang Wang, **Xianfei Wang**, and Xia Hou. "UPSR: a Unified Proxy Skeleton Retargeting Method for Heterogeneous Avatar Animation." 2023, pp. 867-868. DOI

† *Undergraduate thesis supervisor.*

MANUSCRIPTS UNDER REVIEW

Xianfei Wang, Xingze Chen, Kani Chen, and Kai Chen. "EvoScreen: Explainable Video Reports for Community Autism Screening." *Submitted to ICSE 2027, Software Engineering in Society (SEIS) Track. Under review.*

Xianfei Wang+, Xingze Chen+, Junhan Wang, Xilin Tao, Bowen Gong, Kani Chen, and Kai Chen. "ASD Super Agent: a tool-orchestrated multi-expert retrieval-augmented clinical advisory system integrating pose-based video screening for autism spectrum disorder." *Submitted to npj Mental Health Research. Under review.*

+ *Equal contribution (co-first authors).*

RESEARCH AND PROJECT EXPERIENCE

ASD Super Agent and EvoScreen
Graduate research project | HKUST Nov 2025 - Present

- Developed a full-stack application for **autism caregiver support**, integrating retrieval-augmented specialist agents, structured assessments, and longitudinal child records.
- Implemented **tool-based orchestration** and parallel consultation across four specialist agents, with contextual memory and dynamic workflows.
- Android AR virtual advisor using **ARCore**, with plane detection, anchored **VRM avatars**, interactive placement and scaling, and speech-driven animation.
- Cowork mode enabling agents to read local case files and save reports in authorized folders, with access controls and overwrite backups (works like Claude Code/Codex).
- **Video-observation** workflow combining movement analysis, acoustic events, and speech transcripts into timestamped evidence for LLM-generated reports, linking observations to video playback.

HKUST Campus Atlas
Interactive 3D campus exploration and conversational wayfinding Sep 2026 - Present

- Built a 3D campus model in Three.js, React, and TypeScript using OpenStreetMap geometry, terrain data, and photo-informed architectural details.
- Implemented multilevel Dijkstra **navigation routing**, including bridge, stair, and elevator connections.
- Integrated bilingual place search and validated LLM tool calls with text and push-to-talk interfaces for scene exploration and route requests.

SysMocap
Creator and developer | Open-source undergraduate thesis project Feb - Jun 2022

- Developed a cross-platform desktop application for monocular-video avatar animation using MediaPipe, Kalidokit, Three.js, Electron, and Vue.
- Implemented skeleton mapping and coordinate transformations for heterogeneous VRM, FBX, and glTF/GLB avatars, with facial, hand, and body animation, and OBS streaming.
- VR/AR live broadcast with **WebXR**.
- Open-source release with **3.2k+ GitHub stars** (October 2026). Repository

INDUSTRY EXPERIENCE

Bauhinia AI
R&D Engineer May 2024 - Aug 2025
HKUST-incubated startup founded by Prof. Kani Chen.

- Designed the web and client interfaces for WeTips; developed all frontend/client-side functionality and the AI-generated reply feature.
- Contributed to Aivilization, a multi-agent social simulation platform, developing registration and invitation workflows and a character facial-expression generation system.

Tencent AI Lab
R&D Intern | Digital Human Team, TEG Feb - Aug 2023

- Contributed to speech and lip-motion generation, processed MetaHuman and ARKit motion sequences, and retargeted avatar animation with visualization in Unreal Engine and Maya.

Beijing Baiyang Technology
Full-Stack Developer Jun - Sep 2022

- Developed web and mobile dance-learning features using MediaPipe Holistic for real-time motion capture, integrating movement scoring and instructional feedback.

Beijing Zhifang Technology
C++ Development Intern Mar - May 2022

- Implemented graphics and pose-computation utilities in C++ and Eigen, including rotation-matrix and quaternion conversions for Unreal Engine projects.

TEACHING EXPERIENCE

Teaching Assistant, HKUST | COMP 4521 Mobile Application Development Fall 2026
Teaching Assistant, HKUST | COMP 2011 Programming with C++ Spring 2026

TECHNICAL SKILLS

- **Programming:** TypeScript / JavaScript, Python, C++; working knowledge of Java and Kotlin.
- **Interactive systems:** React, Next.js, Vue, Three.js, Electron, MediaPipe, Unreal Engine, Maya.
- **Backend and tools:** FastAPI, Node.js / Express, SQLite, MongoDB, ChromaDB, Git, Linux.

SELECTED HONORS

Beijing Outstanding Undergraduate Thesis; Imagine Cup Beijing Regional Second Prize; Gold Award, Capital Challenge Cup Entrepreneurship Competition.